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The How and the Why of Study Choice Processes in Higher Education: The Role of Parental Involvement and the Experience of Having an Authentic Inner Compass

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Abstract

Late adolescents differ in the degree to which they are thoroughly engaged in the study choice process and in the degree to which their choices are autonomous in nature. This study examined the unique and interactive roles of (a) parental involvement in the study choice process and (b) late adolescents' sense of having an authentic inner compass (AIC) in predicting their study choice decision-making. A cross-sectional study among 331 12th-grade adolescents (M_{age} = 18.04, SD = .48) revealed that late adolescents experiencing a stronger AIC and more need-supportive parental involvement showed more engagement in and autonomous regulation of the study choice process. In contrast, when experiencing more controlling parental involvement or uninvolvement, late adolescents showed more controlled regulation, with parental control also being linked to less commitment. Although mothers were perceived to be more involved than fathers, maternal and paternal involvement were equally strongly related to the study choice tasks.

Keywords: Study Choice Processes, Parental involvement, Self-Determination Theory, Identity, Adolescence

For most late adolescents, graduation from high school marks an important transition involving the decision whether they embark on higher education studies (or not) and, if so, which study they will choose (Glorieux et al., 2014). These decisions can be challenging and stressful, with some adolescents being indecisive (Germeijs et al., 2006) or feeling pressured in an unwanted direction (Hegna & Smette, 2017). Others complete this decision-making process more easily, fully engaging in the study choice process and arriving at a solid commitment to a study choice (Germeijs et al., 2012). Both adolescents' type of motivation for their plans after secondary education and their engagement in the study choice tasks have been found to predict their adjustment to the higher education context, with more autonomous motivation and a more thorough engagement in the decision-making process relating to better adjustment (e.g., Germeijs & Verschueren, 2006; Yu et al., 2018). Therefore, it is essential to identify resources and risk factors that predict adolescents' motivation for and way of dealing with this decision-making process.

Study choice processes are considered to be a function of both contextual and personal resources (Hirschi et al., 2011; Holmegaard et al., 2014). Parents in particular are assumed to play a critical role, either supporting or complicating adolescents' study choice process (Dietrich & Kracke, 2009). In terms of personal resources, it has been argued that it is crucial for adolescents to experience a strong inner foundation, referred to as an authentic inner compass (AIC; Assor, 2018), that provides direction and confidence to engage in the process of decision-making. Although studies have begun to document either the role of parental involvement or the sense of having an AIC in adolescents' study choice decision-making, no research to date examined the unique and interactive roles of both factors simultaneously. This is unfortunate because the relative contribution of the parental context and late adolescents' personal resources to the study choice process remains unclear. Also, no research to date addressed whether a personal resource (i.e., the sense of having an AIC) can buffer against a lack of constructive parental involvement. These questions are also important from an applied perspective because they can inform researchers and practitioners about the most important target for interventions and individual counseling efforts to strengthen late adolescents' study choice process.

Higher Education Study Choice Process

Late adolescents face the task of developing their identity in different life domains (e.g., relational, political, vocational), and choosing what to study reflects one crucial decision within this broader identity process. Much research has focused on *how* late adolescents tackle the complex task of making a higher education study choice. Germeijs and Verschueren (2006) differentiated between six decisional tasks in the context of the higher education study choice process. The first task is that late adolescents need to *orient* themselves to the study choice task. That is, they become increasingly aware that making a decision is necessary and feel motivated to engage in the process. Next, late adolescents need to engage in an exploration process differentiated by three decisional subtasks. Specifically, adolescents gather (a) information about their interests, values, abilities, and study strategies (i.e., *self-exploration*), (b) general information about the different study programs (i.e., *broad exploration of the environment*), and (c) more detailed information about a limited amount of alternatives (i.e., *in-depth exploration of the environment*). A fifth task is that late adolescents proceed to a *decision* about which major they will follow. Finally, they eventually *commit* to their chosen major, as indicated by a strong attachment to the chosen alternative.

Although these distinct decisional tasks can be performed cyclically, they typically build up in a gradual order, with orientation and broad exploration occurring mainly at the beginning of the process, followed by exploration in depth and, finally, in the ideal scenario, a solid commitment with the study choice (Germeijs & Verschueren, 2006). A thorough engagement in these different decisional tasks has been shown to be vitally important for students' later academic functioning (Germeijs et al., 2012). For instance, Belgian first-year university students who reported having made an informed choice, that is, a choice made after exploration of different possible educational programs and discussions with others, obtained better grades at the end of the academic year (De Clercq et al., 2017). In a prospective study, Germeijs and Verschueren (2007) showed that late adolescents who reported more exploration and initial commitment to the study choice in Grade 12 were more committed to the chosen study and showed better adjustment to the academic climate once in college or university.

Apart from how late adolescents approach this decisional task, an equally important question is *why* they decide to embark on higher education studies and why they choose a particular study (e.g., Guay, 2005). One highly influential framework in research on this motivational regulation of study choices is Self-Determination Theory (SDT; Ryan & Deci, 2017). This theory distinguishes between more controlled (pressured) or more autonomous (volitional) reasons for making important and identity-relevant choices (Ryan & Connell, 1989; Ryan & Deci, 2003; Soenens & Vansteenkiste, 2011). When being *controlled motivated*, late adolescents experience pressure to pursue a future goal. This pressure can be more external, which is the case when significant others compel the adolescent to follow a particular career path. However, pressure can also originate from within when students pursue a goal to avoid guilt or anxiety or to feel proud of themselves (i.e., introjection). In contrast, when being *autonomously motivated*, late adolescents see the personal value and relevance of a particular decision (identification) or even make this choice based on curiosity and anticipated interest and enjoyment (intrinsic motivation).

An important assumption in SDT is that autonomous motivation contributes to better adjustment and performance than controlled motivation (Ryan & Deci, 2017). Although this assumption has received much empirical support in diverse life domains (including education, exercise, and work; e.g., Vasconcellos et al., 2020), only a limited number of studies to date focused specifically on students' motivational regulation of their study choices and future plans as such. In a large sample of US university students, Soria and Stebleton (2013) found that autonomous motivation to choose a specific major was predictive of more educational satisfaction and a greater sense of belonging to the university. In contrast, controlled motivation was associated with less satisfaction and belonging. Similarly, a study with a mixed sample of American and Chinese college students showed that autonomous motivation to choose a study major was positively related to students' well-being and actual performance (Yu et al., 2018). Research from the Netherlands confirmed that bachelor students who made study choices for more autonomous reasons performed better and were less likely to drop out in the first year of higher education (Meens et al., 2018).

Given the multiple beneficial outcomes associated with a thorough engagement in the study choice decision-making process (i.e., the how of decision-making) and with an autonomous regulation of study choices (i.e., the why of decision-making), it is crucial to shed more light on the antecedents of these two facets of the decision-making process. Much research has been conducted from a sociological viewpoint, pointing toward students' socio-demographic background characteristics (e.g., gender, socioeconomic status; Bergerson, 2009; Reay et al., 2005). However, drawing from transactional models emphasizing the interactive effects of parenting and adolescent characteristics in predicting adolescent development (Kiff et al., 2011; Sameroff, 2009), we focus on the roles of parental involvement and late adolescents' sense of having an AIC as a contextual and individual resource, respectively, for the study choice process.

The Role of Parental Involvement

Parents' involvement in their children's school functioning plays a vital role in children's educational outcomes and academic trajectories (e.g., Katz et al., 2018; Vasquez et al., 2016; Wilder, 2014). Considering the importance of late adolescents' study choice process, the question arises whether and how parents can support, complicate or even interfere in this process.

Research already showed that both general relationship quality (e.g., attachment, Germeijs & Verschueren, 2009) and study- and career-specific parental involvement (e.g., Hirschi et al., 2011; Holmegaard et al., 2014; Mao, 2014; Mortimer et al., 2002) are predictive of adolescents' engagement in this decision-making process. For instance, Dietrich and Kracke (2009) showed that adolescents who perceived their parents as supportively involved in their study decision-making process reported more career exploration behaviors. In contrast, adolescents with parents who lacked involvement reported more career exploration difficulties. A longitudinal study by Guan et al. (2015) similarly showed that adolescents with involved parents engaged more often in career exploration behaviors, which in turn predicted confidence in their coping with career-related problems.

Importantly, this emerging research shows that parents' involvement style is important and may even matter more than the degree of parental involvement per se (e.g., Dietrich & Kracke, 2009; Guan et al., 2015). We draw from Self Determination Theory (SDT; Ryan & Deci, 2017) to differentiate between a need supportive style of being involved in career decision-making, a more controlling style of parental involvement, and parental uninvolvement.

When parents are involved in a *need supportive* way, they give adequate guidance and support (i.e., providing structure) while at the same time acknowledging the adolescent's perspective and encouraging the adolescent to make personal decisions based on their own values and interests (i.e., autonomy-support; Grolnick et al., 2015; Skinner et al., 2005; Deci, & Ryan, 2017). In contrast, parents can be more *controlling* in their involvement. Controlling parents prioritize their own ideas, for instance, by making their appreciation and regard dependent upon their adolescents' career choices, using fear tactics or bluntly enforcing the adolescent to comply with their preferred study choice. Parents can also be generally *uninvolved* in the career decision-making process. They then leave their adolescent at their own device and provide little to no help, expectations, or feedback (Skinner et al., 2005). This lack of involvement can emerge either from parental (over)confidence in the adolescent's ability to decide independently or from of a lack of time or interest to invest in this process.

In general, need-supportive parenting has been found to relate to positive indicators of academic functioning (Vasquez et al., 2016), including motivation, engagement, and achievement (e.g., Grolnick et al., 2015; Katz et al., 2011; Roth et al., 2009; Soenens & Vansteenkiste, 2005). In contrast, controlling parenting predicts more problematic academic outcomes, such as less engagement (Fei-Yin Ng et al., 2004), lower performance (Grolnick et al., 2002), academic procrastination (Won & Yu, 2018), and a decreased academic self-concept (Calders et al., 2020).

A few studies have also begun to examine the role of a need supportive parenting style in the study choice process (e.g., Ahn et al., 2022; Cordeiro et al., 2018; Soenens & Vansteenkiste, 2005). For example, Guay et al. (2003) demonstrated that parental need support was linked with feelings of autonomy and self-efficacy in the study choice process and related negatively to study choice indecision. Parental need support was even found to predict students' persistence in the chosen study three years after enrollment (Ratelle et al., 2005). Katz et al. (2018) showed that perceived need supportive parentinging related positively to adolescents' sense of self-determination towards choosing a major subject, which in turn predicted more autonomous motivation to learn and better grades. Research concerning the role of controlling parenting or uninvolvement in the study choice process is much sparser. However, the conducted studies did observe in general that parental control and uninvolvement was linked with less exploration in the decision-making process by the adolescent and more difficulties in choosing a study (Ahn et al., 2022; Cordeiro et al., 2018; Dietrich & Kracke, 2009; Guan et al., 2015).

One important limitation in this emerging research on parental involvement in the study choice process is that only few studies differentiated between mothers' and fathers' involvement. Whereas some studies focused on mothers only, others relied on general assessments of parents' behavior not differentiated by parental gender. This lack of differentiation between maternal and paternal involvement is unfortunate because research has documented mean-level differences in parenting styles and their relation to adolescent outcomes, concluding that more gender-sensitive research in parenting in general (Schiffrin et al., 2019b) and parental involvement in the study choice process (Ahn et al., 2022) is needed. Indeed, previous studies observed, for instance, that fathers tend to be less controlling than mothers (Schiffrin et al., 2019a), and that children report less internalizing and externalizing problems when they have involved fathers, and not mothers, with whom they feel connected (Day & Padilla-Walker, 2009). Concerning study choice decision making, students tend to turn more to their mother to have a conversation about school programs (Muller, 1998; Otto, 2000), especially during moments when they actively explore different study possibilities (Dietrich et al., 2011). In addition, mothers tend to be more autonomy-supportive than fathers when adolescents face study choices (Dietrich et al., 2011; Duineveld et al., 2017), whereas mothers and fathers do not differ regarding the use of control (Dietrich et al., 2011). Moreover, it has been argued by some scholars that fathers could perhaps play a more decisive role in the study choice process because fathers, as so-called 'ministers of foreign affairs', would play a more prominent role encouraging and stimulating children to explore the outer world compared to mothers, who would serve more often as 'ministers of internal affairs', thereby primarily playing a role as a safe haven in times of distress (Grossmann et al., 2008; Verschueren & Marcoen, 1999).

To date, a few studies provided indirect evidence that paternal support plays a relatively more substantial role in career exploration behaviors than maternal support. For example, Beyers and Goossens (2008) showed that paternal support was related more strongly to broad identity exploration in adolescents than maternal support. Similarly, Soenens and Vansteenkiste (2005) showed that paternal autonomy-support, but not maternal autonomy-support, was related to more autonomous motivation for job search behaviors which, in turn related to more vocational exploration and commitment among late adolescents. In contrast, other studies with adolescents found that maternal support related more to career development than paternal support (Ginevra et al., 2013) and Germeijs and Verschueren (2009) observed that only the quality of attachment with mother and not father was predictive for late adolescents' study choice process.

Given the conflicting findings regarding maternal and paternal involvement in study and career development of adolescents, additional research is needed to more directly examine the unique and combined effects of maternal and paternal involvement in late adolescents' study choice decision-making. Moreover, based on transactional models of development emphasizing the need to study interactions between individual characteristics and environmental factors to fully understand developmental processes (Kiff et al., 2011), it is essential to consider the role of parental involvement in combination with personal resources for decision-making. One such resource with much importance for decision-making processes is the degree to which adolescents experience having an authentic inner compass.

The Role of Adolescents' Sense of Having an Authentic Inner Compass

According to SDT (Ryan & Deci, 2017), adolescents ideally make important identity-relevant choices such as a study or a career based on a firmly anchored and autonomously regulated identity (Soenens & Vansteenkiste, 2011; Meens et al., 2018). Assor (2012, 2017; Assor et al., 2020) argued that identity commitments are more autonomous and personally expressive when they reflect individuals' *Authentic Inner Compass* (AIC). This AIC refers to the experience of having a set of clear and authentic values, goals, and interests that informs people about what they truly value and need, thereby guiding decision-making. Such values, goals, and interests are considered authentic when they are experienced as being free from pressure, and growth-oriented, that is, directed towards the satisfaction of one's need for autonomy (Assor et al., 2023).

Consistent with this reasoning, several recent studies have shown that adolescents differ in the extent to which they experience having an AIC and that these individual differences matter for the quality of adolescents' identity development and decision-making. For instance, Russo-Netzer & Shoshani (2020) have shown that adolescents with a strong sense of AIC deliberately make more choices that contribute to their sense of meaning and positivity. Furthermore, a strong sense of AIC was associated with choosing to resist peer pressure (Assor, Benita et al., 2020). Moreover, Assor et al. (2020) observed a positive association between the sense of having an AIC and autonomous identity commitments in high school students. Adolescents who feel more in touch with their most fundamental values, interests, and preferences, appear to be better able to arrive at personally endorsed identity commitments, that is, commitments they truly identify with. In turn, such autonomously regulated identity commitments have been found to contribute to positive outcomes, including well-being (e.g., Sheldon & Kasser, 1995; Waterman et al., 2013) and academic adjustment (e.g., Luyckx, 2005).

Based on this emerging research, it can be assumed that adolescents' sense of having an AIC facilitates study choice processes in the transition to higher education. When confronted with a large array of possible studies, the AIC provides self-direction to make decisions conducive to their preferences and needs (Assor et al., 2023). The sense of having an AIC provides an internal criterium to explore options in greater depth and in a more structured, thorough, and purposeful way. The sense of having an AIC also helps to make a committed decision, because it would be more directly clear to people which decision matches their interests, preferences, and goals. Late adolescents would not necessarily have to go through an elaborate and cognitively mediated process of deliberation and would instead often almost intuitively experience a fit (or mismatch) between a study choice option and their inner characteristics. In addition, a strong sense of AIC could provide motivational resources to engage in this process, as the study choice process would be perceived more as an opportunity to develop further rather than as a burden or a threat. The current study is among the first to examine this assumption, thereby contributing to the empirical examination of the notion of an AIC, a construct that provides conceptual depth but that has received limited empircical attention to date. Moreover, it also extends previous research by examining the role of the AIC, as a personal resource for decision-making, combined with parental involvement, as a contextual resource. As such, in line with a transactional view on adolescent development (Sameroff, 2009), the present study will address both the unique predictive value of these two resources and the possible interactive interplay between late adolescents' sense of having an AIC and parental involvement. To our knowledge, no research has examined this issue yet. However, shedding light on this matter could be of theoretical and practical value. It could be for instance that having a sense of AIC and parental involvement play an independent role in predicting higher education study choice decision making, with one factor having a stronger predictive value. From a practical point of view, this would imply that, when confronted with an adolescent who experiences trouble choosing a study, intervention efforts should target both predictive factors, with more emphasis on the one that has the strongest predictive value. On the other hand, it could be that these two resources can have compensatory effects on late adolescents' decision-making tasks. Specifically, late adolescents with a strong sense of having an AIC may need less parental support for the decision-making process, thereby being less affected in their study choice process by how parents are involved. These adolescents may make identity-relevant choices more independently and based primarily on their experience of having an AIC. Conversely, constructive parental involvement may be needed the most when adolescents lack a sense of having a sense of an AIC. Parental involvement may then be required to provide guidance and facilitate adequate decision-making in the absence of the direction provided when experiencing a firm sense of AIC and intervention programs could focus on this contextual factor in order to compensate for the lack of a firm sense of AIC of the adolescent.

The Present Study

The main aim of the present study was to examine the unique associations between parental involvement and the AIC in relation to late adolescents' study choice process. Based upon previous research (e.g., Assor et al., 2020; Dietrich & Kracke, 2009; Guan et al., 2003, 2015), it was hypothesized that need-supportive parental involvement and the experience of having an AIC would be associated positively with adolescents' engagement in the study choice tasks (i.e., the "how" of the study choice process) and with autonomous motivation for future plans while being negatively related to controlled motivation for future plans (the "why" of the study choice process). The opposite pattern of associations was hypothesized for a controlling and uninvolved parental involvement.

Two additional research questions were addressed in this study, which are, given the limited research available and the conflicting findings, exploratory in nature. That is, first, in light of previous research showing parental gender differences in the level of involvement (e.g., Dietrich et al., 2011; Duineveld et al., 2017) and the relation with adolescents' identity development (Beyers & Goossens, 2008), both maternal and paternal involvement in the study choice process of late adolescents were examined. This approach allowed us to examine mean-level differences in parental involvement by parental gender and to address both the combined and differential effects of maternal and paternal involvement. Second, we explored the possible interplay between the AIC and parental involvement in the prediction of late adolescents' study choice process.

Method

Participants and Procedure

The sample for this study was collected in [details removed for peer review], and comprised 331 of last year's high school students (68.3% girls, M_{age} = 18.04, SD = 0.48). The majority of the participants followed an academic track (71.1%), with other participants following a technical (22.7%) or vocational (6.2%) track. Most participants lived with both parents in an intact family (78.5%), and 4.8% lived in a co-parenting agreement. A minority of adolescents lived solely with the mother (14.5%) or the father (1.2%). Regarding parental education, 64.2% of the fathers (33.3% bachelor, 30.9% master's) and 74.8% of the mothers (49.2% bachelor, 25.6% master's) had a higher education degree.

All participants received paper-and-pencil questionnaires in the Spring of 2017 and 2018. These questionnaires were administered in school. Before filling out the questionnaires, active informed consent was obtained from the adolescents, whereas passive informed consent was obtained from their parents. The study was conducted according to the ethical rules presented in the General Ethical Protocol of [details removed for peer review].

Measures

Study Choice Process. A short version (Demulder et al., 2020) of the Study Choice Task Inventory (SCTI; Germeijs & Verschueren, 2006) was administered to assess the higher education study choice process. The first subscale, *Orientation to Choice,* measures the degree of awareness and willingness to engage in the study choice process (e.g., "I want to do my best to make a good study choice", 7 items). The items from this scale were rated on a 5-point Likert scale from 1 *totally not applicable* to 5 *totally applicable*. Next, three subscales were used to measure the degree of exploration, that is, the *Self-Exploratory Behaviour* scale (i.e., exploration of their abilities, interests, values, and study skills; e.g., "I have thought about what I like and don't like doing", 8 items), the *Broad Exploratory Behaviour* scale (i.e., general exploration of different study programs; e.g., "I have looked at leaflets and websites from different fields of study", 5 items) and the *In-depth Exploratory Behaviour* scale (i.e., thorough in detail exploration of a limited amount of study programs; e.g., "I have talked to students who are now in higher education about one of the study programs", 7 items). The items from these three subscales were rated on a 4-point Likert scale ranging from 1 *never* to 4 *very often*. Using an aggregated score for exploration is justified by the positive correlation between the three subscales (.41 < r < .60). Because Germeijs and Verschueren (2007) argued that exploration (the process of gathering information) must be distinguished from the actual information that is present, we also administered a subscale measuring *Information* (8 items; Germeijs et al., 2006) which aims to assess the amount of information students have about themselves and their environment, going from 1 *completely disagree* to 5 *completely agree*. A sample item is "I know which courses included in the study programs interest me". Finally, the *Commitment* scale measures the degree to which students identify with the preliminary study choice (e.g., "Are you sure of your choice for this field of study?", 6 items). These items were rated on a Likert scale going from 1 *definitely not* to 6 *definitely*. The SCTI has found to be a reliable and valid questionnaire to measure study choice decision-making (e.g., Germeijs et al., 2012) and for the short version that we used, internal consistencies in our sample were excellent for orientation to choice (α = .82), exploration (α = .90), information (α = .80) and commitment (α = .85).

Motivation for Future Plans. We measured students' autonomous and controlled motivation for their future plans using a slightly adapted version of the validated and theoretically grounded Motivation for Identity Commitments scale (Soenens et al., 2011). Participants were first asked to indicate their concrete plan for the future, thereby making a selection from seven possible plans, four of which involved a study choice (e.g., "I want to attend higher education, and I am considering several study programs"), whereas the others involved going to work, taking a sabbatical or not having a future plan yet. Then, preceded by the item stem "I pursue this plan because…", students were asked to rate items reflecting their motivation for this future plan on a scale ranging from 1 *totally not* to 5 *totally*. Sample items included "… because I think this plan is meaningful for myself" (*autonomous motivation*, 4 items, α = .83) and "…because I am obliged to do so by others" (*controlled motivation*, 4 items, α = .80).

Perceived Parental Involvement. To assess perceived parental involvement in the study choice process, participants filled out three subscales, rating items separately for mothers and fathers and using a 5-point scale ranging between 1 (*totally not*) and 5 (*totally*). Two of the three subscales were selected from the widely used (e.g., Zhang et al., 2019) Scale of Perceived Parental Career-related Behaviours (Dietrich & Kracke, 2009). The items were adapted to fit the study choice context, measuring parental interference or *control* (5 items; e.g., "…tries to push my study choice in a certain direction"; α_{mothers} = .79, α_{fathers} = .82) and lack of engagement or *uninvolvement* (5 items; e.g., "…is not really interested in my future study or profession"; α_{mothers} = .69, α_{fathers} = .75). Based on SDT, we added a scale with 13 items to measure *parental need support.* The items reflected both autonomy-supportive (e.g., "…listens to my opinion about my future study or profession") and structuring practices (e.g., "…helps me figure out which study and career choices exist"). Based on a factor analysis, one item (i.e., "…gives me the freedom to choose my own study or profession") was removed from this scale because it had a low loading. The internal consistencies for parental need support were excellent (α_{mothers} = .92, α_{fathers} = .94).

Authentic Inner Compass (AIC). To assess adolescents' sense of having an AIC, we relied on the [details removed for peer review] translation (Assor et al., 2020) of the AIC Foundation scale (Assor, 2018). This 8-item scale (e.g., "I have goals that truly reflect the person I want to be") has been shown to be reliable and valid in previous research (Assor, 2020). In the present sample, internal consistency was satisfactory ($\alpha = .77$).

Plan of Analysis

First, to examine whether mothers differ from fathers about the nature of their involvement in the study choice process, three paired samples *t*-tests were performed. Next, a latent sum and difference model (Nelson et al., 2006) was estimated using Mplus 8.4 (Muthén & Muthén, 1998-2017) to estimate both the combined and unique effects of maternal and paternal involvement (e.g., Kuppens et al., 2009). First, a measurement model was fit with three latent factors for both maternal and paternal involvement (i.e., need support, control, and uninvolvement). Each of these factors were indicated by three parcels, the loadings of which were fixed across parental gender. All parcels comprised the same items for mothers and fathers. These latent factors for the maternal and paternal practices were used as indicators for second-order latent factors representing the common (i.e., the cumulative effect of both parents) and differential (i.e., unique and different effect for both parents) effects for each style of parental involvement. Concerning the differential effects, positive coefficients represented an effect stronger for mothers, whereas negative coefficients represented an effect stronger for fathers. For each style of involvement, the common and differential effects were unrelated ($.33 < p < .66$). Model fit was assessed based on the combined cut-off criteria provided by Hu and Bentler (1999): CFI > .90, RMSEA < .06, and SRMR < .08. After allowing three additional error correlations between similar parcels in the maternal and paternal ratings, the measurement model showed an acceptable fit ($\chi^2(123) = 313.27$, CFI = .93, RMSEA = .07, SRMR = .06). Second, for each outcome variable separately, a latent sum and difference model was estimated with both the common and differential effects of parental involvement (i.e., need support, control, uninvolvement) as predictors of the study choice process (i.e., orientation, exploration, information, commitment, autonomous motivation, and controlled motivation). Third, in the final and most comprehensive model, we entered the common effects of parental involvement together with late adolescents' AIC as predictors. To test the moderating role of AIC in the relation between parental involvement and the study choice, we entered the interaction term between each of the common factors of parental involvement and the AIC.

Results

Descriptive Statistics and Preliminary Analyses

First, as part of the data was missing (4.6%), Little's (1988) MCAR test was performed, showing that data was most likely missing at random (i.e., normed χ^2 (1573.67/1301) = 1.21; Ullman, 2001). Therefore, missing values were estimated using the EM-algorithm (Schafer, 1997).

Second, in order to control for relevant background variables in the structural analyses, a MANCOVA was conducted to examine whether participants' personal (i.e., age, gender), educational (i.e., educational track, past test results) and family background characteristics (i.e., living situation and parental education level) were related to any of our study variables. Given the relatively large sample size, significance level for all performed analyses was set to $p < .01$ in order to avoid Type I errors. Results indicated that there was no significant multivariate effect for adolescents' age, living situation and parents' educational level (Wilks's λ ranging from 0.77 to 0.97 and all $p > .01$). We did find significant multivariate effects of adolescents' gender (Wilks's $\lambda = 0.83$, $F(13,161) = 2.60$, $p < .01$, $\eta^2 = .17$) and educational track (Wilks's $\lambda = 0.72$, $F(26,322) = 2.26$, $p < .01$, $\eta^2 = .15$). Univariate follow-up analyses revealed that male adolescents scored lower than female adolescents on orientation ($M_{male} = 3.40$, $SD = 0.16$; $M_{female} = 3.79$, $SD = 0.14$; $F(1,173) = 11.04$, $p < .01$, $\eta2 = .06$) and exploration ($M_{male} = 2.54$, $SD = 0.12$; $M_{female} = 2.82$, $SD = 0.11$; $F(1,173) = 8.95$, $p < .01$, $\eta2 = .05$). Futher, students following an academic track reported less commitment with their study choice ($M = 4.59$, $SD = 0.12$) than students following a technical ($M = 5.12$, $SD = 0.20$) or vocational track ($M = 5.59$, $SD = 0.49$; $F(2,173) = 4.82$, $p < .01$, $\eta2 = .05$), whereas both students following an academic ($M = 2.04$, $SD = 0.10$) and vocational track ($M = 2.77$, $SD = 0.40$) reported more controlling maternal involvement compared to students following a technical track ($M = 1.73$, $SD = 0.16$;

$F(2,173) = 6.12$, $p < .01$, $\eta 2 = .07$). As a result, we controlled for adolescents' gender and educational track in the main analyses.

Table 1 displays the descriptive statistics and bivariate correlations among the measured variables, with these correlations being controlled for the significant background variables. As seen in this table, parental need support was related positively to autonomous motivation, orientation, exploration, and information, whereas parental control was linked to more controlled motivation. In addition, paternal (but not maternal) control was related negatively to information and commitment. Parental uninvolvement was related positively to controlled motivation and negatively to autonomous motivation, information, and commitment. Maternal (but not paternal) uninvolvement was associated with less exploration. Finally, adolescents' sense of having an AIC was related positively to all outcome measures except for controlled motivation, to which it was related negatively.

Main Analyses

Parental Involvement in the Study Choice Decision Process. With regards to mean-level differences in parental involvement by parental gender, results of the paired samples *t*-tests showed that mothers are perceived by their adolescent as less uninvolved ($M_{mother} = 1.49$, $SD = 0.60$; $M_{father} = 1.88$, $SD = 0.82$; $t(306) = -8.31$, $p < .001$, $d = 0.65$), higher on need supportive involvement ($M_{mother} = 3.90$, $SD = 0.73$; $M_{father} = 3.37$, $SD = 0.88$; $t(305) = 9.83$, $p < .001$; $d = 0.60$) and higher on controlling involvement ($M_{mother} = 2.26$, $SD = 0.83$; $M_{father} = 2.02$, $SD = 0.88$; $t(306) = 4.13$, $p < .001$; $d = 0.27$). Apparently then, mothers were perceived to be more involved overall.

Next, the latent sum and difference models estimating common and differential variance in parental involvement on the study choice tasks fitted the data well (CFI = .92 to .93, RMSEA = .06 to .07, SRMR = .06). The common factors of parental involvement on average captured 69.4% of the variance of the mother and father scores, whereas the differential factors captured 30.6% of this variance. Relationships between common and differential parental involvement and study choice variables are displayed in Table 2. The extent to which mothers and fathers are both need supportive (i.e., their common involvement) was associated with more study orientation and exploration. In contrast, the degree to which mothers and fathers are both uninvolved or controlling was related to more controlled motivation, with common control being associated with less commitment as well. None of the differential effects that were estimated were significant predictors of the study choice variables.

The Unique and Interactive Role of the Authentic Inner Compass. The final model included the simultaneous effects of parental involvement and adolescents' sense of AIC, as well as their interactions. Because the differential effects were unrelated to the study choice decision-making and because our primary interest was in the unique roles of parents' overall involvement and the sense of AIC, we only included the common factors for the parental involvement variables in these models. As Table 2 shows, late adolescents' AIC predicted all outcomes, that is, more autonomous motivation, less controlled motivation, more orientation, exploration, information, and commitment. Further, all but one (i.e., the common effect of need support on study orientation) of the main effects of parental involvement remained significant after adding the AIC. Only one of the 18 interaction effects between parental involvement and the sense of AIC was significant.

Discussion

Although considered a time of opportunity and growth, the transition to the first year in higher education can be demanding (Briggs et al., 2012), as it often requires late adolescents to move to a different city, to build up a new peer network, and to handle higher educational demands and societal expectations (Duineveld et al., 2017). Relying on research showing that the way students engage in the study choice process is predictive of their subsequent academic functioning (e.g., Germeijs & Verschueren, 2007), the present study attempted to gain insight into the contextual (i.e., parental involvement) and personal (i.e., late adolescents' authentic inner compass) predictors of adolescents' engagement in the study choice process and their autonomous motivation for this higher education study choice.

The Need for a Supportive Maternal and Paternal Context

The present study provides evidence that late adolescents may benefit from parental involvement if at least the involvement is need supportive rather than controlling. Indeed, in line with SDT (Ryan & Deci, 2017), the degree to which parents are both need supportive in their involvement is associated with adolescents' greater awareness and orientation towards the study choice and more exploratory behavior, including a broader and more in-depth comparison of different study programs and the self. These results coincide partly with Cordeiro et al. (2018), who found that more general parental support was related to more exploration and commitment. When parents were perceived to be more uninvolved or controlling, late adolescents reported less engagement in the study choice tasks. Moreover, late adolescents who experienced their parents as more controlling felt less certain about their preliminary study choice and experienced more pressure to make a particular choice (i.e., controlled motivation). Adolescents' degree of exploration did not differ depending on whether parents were perceived to be uninvolved or involved in a controlling way. This contrasts with Dietrich et al. (2011), who found elevated exploration when parents, particularly mothers, were controllingly involved. One possible explanation for these divergent findings is that adolescents' exploration can be regulated by more autonomous or controlled motives (Smits et al., 2010), with controlling parenting perhaps being particularly predictive of a controlled motivation for exploration. Future research would indeed do well to further examine whether need-supportive involvement is related particularly to a high-quality motivational regulation of exploratory behavior, with adolescents engaging in exploration because they want to and whether controlling involvement relates to adolescents' tendency to explore options just because they have to.

Although common effects for parental involvement were present for all but two of the outcomes (i.e., autonomous motivation and information), differential parent effects were absent. This finding indicates that the degree to which mothers and fathers are similarly involved plays a more decisive role in late adolescents' decision-making processes than the degree to which mothers or fathers displayed elevated involvement compared to the other parent. Although somewhat inconsistent with previous theorizing suggesting that fathers may play a more prominent role in career and study choice decision-making (e.g., Beyers & Goossens, 2008; Grossmann et al., 2008; but see Germeijs and Verschueren (2009) for an exception), the current findings converge with Dietrich et al. (2011), who also found more similarities than differences between effects of involvement across parental gender.

Because the present study was among the first to examine the combined and unique effects of maternal and paternal involvement in late adolescents' study choice decision-making, more research is needed. It should also be noted that our data did indicate significant mean-level differences in parental involvement by parental gender. Overall, mothers were more involved in the study choice decisions of their child, both in need supportive and controlling ways. This higher overall level of involvement is consistent with the observation that, although fathers are gradually becoming more involved in children's lives in recent years, mothers are still generally more involved than fathers (Nomaguchi & Milkie, 2020). This broader literature on differences in parental involvement by parental gender has also shown that mothers are more involved in life domains and tasks that require much mental energy, such as the planning of children's social and extracurricular lives (Bianchi et al., 2006). Clearly, study and career decision-making processes involve a high cognitive load, and mothers' greater engagement in these tasks than fathers meshes with the observation that mothers spend more mental energy when involved with their children than fathers.

A Sense of Having an Authentic Inner Compass as a Guidepost for Decision-Making

While the present study and previous research have shown that parents can support or interfere with the study choice process of late adolescents, the current results also indicate that regardless of how parents are involved, late adolescents benefit from experiencing an AIC. Adolescents who have a clear view of what values, goals and interests they find truly important in life, were found to be more engaged in their search for a suitable study and to possess more information to guide that decision-making. They additionally arrived more easily at plans for the future that they experienced as personally meaningful and exciting. Indeed, information and autonomous motivation were only predicted by late adolescents' sense of an AIC (and not by parental involvement). Although this sense of an AIC seems to serve as an important guidepost for behavior, the present results show an additive effect of parental need support on exploration, parental control and uninvolvement on controlled motivation and parental control on commitment.

The observation that late adolescents' sense of having an AIC was overall the strongest, most systematic, and, in some cases, a unique predictor of their decision-making might be because the sense of AIC is the more proximal predictor compared to parental support. The present study sample consisted of late adolescents in their last year of high school. At this age, many young people already have quite a clear view of who they are and what they find important in life, which is illustrated by the high mean level of AIC in the present sample. In addition, throughout adolescence, the relative influence of parents diminishes as adolescents become more agentic in their life choices and as other contexts (e.g., peers and romantic partners) become gradually more important (De Goede et al., 2009).

These findings indicate that adolescents need a well-established inner sense of self to guide them through the study choice process. From a prevention point of view, we suggest that parents, schools, and counselors are aware of the importance of late adolescents' sense of AIC and try to support that AIC foundation. Before guiding students to go through the different phases of the decision-making process (e.g., gathering information, exploring options, and weighing the benefits and disadvantages of choices), it may be essential to have adolescents make contact with their sense of AIC. By doing so, late adolescents are less likely to drown in the information they gather, more likely explore a limited set of meaningful options (rather than just an overly broad array of all possible choices), and more likely make choices they are deeply committed to. Assor et al. (2020) suggest several practices to support the AIC foundation. Given the increasing pressure to choose a suitable study in the final year of secondary education (Ahn et al., 2022), it may be that especially fostering inner valuing can make the difference for adolescents becacuse this practice entails communicating the importance of resisting social pressure and of taking time to reflect on their AIC when facing a dilemma (Assor et al., 2020). From a remedial perspective, student counselors who help support the study choice process of young people could be advised to consider the broader identity of late adolescents as it could potentially fuel this process. In addition, since the style of parental involvement has an additive effect on adolescents' study choice process, parents can be advised to refrain from controlling tactics to guide study choice decision-making and to provide structure in an autonomy-supportive way, for instance, by discussing the value of engaging in their study choice process (Cheon et al., 2020).

Contrary to our expectations, late adolescents' sense of AIC and their parents' perceived style of involvement were related independently to the decision-making process outcomes. We did not find any evidence for compensatory or synergistic interaction effects. Because the present study is among the first to explore such interactions, more research is needed to still consider a potential interactive interplay between parental involvement and adolescents' sense of AIC. Possibly, such interactions predict changes across time in study choice-decision-making tasks (rather than concurrent between-student differences, as examined in this study) or in earlier phases of this process (e.g., in early and middle adolescence). Future research would also do well to examine other types of interplay between parenting, the sense of AIC, and adolescent outcomes. In contrast to parents' domain-specific involvement, parents' more general parenting, and their engagement in autonomy-supportive communication in particular, may actually contribute to the development of adolescents' sense of AIC over time (Assor et al., 2020). In addition, a more direct active encouragement to reflect upon what they find truly important in life could promote late adolescents' sense of AIC, in turn predicting high-quality decision-making. That is, once a strong sense of AIC is formed and internalized, late adolescents might feel sufficiently armed to tackle important identity choices, as they can rely on strong internal guidance.

Limitations and Future Directions

The present study has various strengths, including the fine-grained measurement of both the study choice process (i.e., how and why late adolescents' engage in this process) and parental involvement (i.e., quantity and quality of parental involvement). However, several limitations are present.

First, this study solely employed adolescents' self-reports, which may cause shared method variance and same-source bias (Podsakoff et al., 2003). It would be informative to use objective measures for some decisional tasks such as exploration, combined with parent-reports of involvement in the study choice process and a qualitative approach to measure identity (e.g., Identity Status Interview; Marcia, 2007).

Second, as this study was cross-sectional in nature, no conclusions about direction of effects (let alone about causality) can be made. Future research would do well to adopt a longitudinal design for at least two reasons. First, a longitudinal design would allow to examine reciprocal relations between the study variables. That is, it could be the case that late adolescents' who show a thorough engagement in the search for a suitable study might evoke a more supportive parental climate, as parents feel that their child is taking this process of decision-making seriously and trust that their adolescent will come to a well-informed decision (Beyers & Goossens, 2008). Dietrich et al. (2011) indeed showed in a longitudinal diary study that parents were more involved when late adolescents were exploring more. Second, a longitudinal design would allow for an investigation of developmental changes in the study variables of this study. Indeed, both studies within the identity literature (e.g., Cordeiro et al., 2015; Luyckx et al., 2013) and studies in the literature on study choice processes (e.g., Germeijs & Verschueren, 2006) identified such developmental changes, pointing towards an adaptive trend over time, although considerable variability between adolescents was present. Future research could therefore examine which factors are most influential at what developmental time. It could for instance be the case that parental involvement is especially important in the beginning of the study choice process, to increase adolescents' awareness of this process and to support an exploration in breadth of different possibilities. When adolescents have eventually narrowed down the different study possibilities, the sense of having an AIC might become especially important, as the choice itself then ideally matches with the adolescent's goals, interests and values. In addition, because not all parents are always and uniquely need-supportive or need-thwarting in their study choice involvement (Ahn et al., 2022; Liang et al., 2022; Zhang et al., 2019), a person-centered approach to parental involvement could help gain a refined understanding of which strategies parents use to support their adolescents and how these strategies might be combined and evolve, for instance, due to time pressure to make a study choice. Ahn et al. (2022) has shown in that regard that controlling strategies sometimes coincide with a more autonomy-supportive parental involvement.

A third limitation is that the current study has a rather selective sample, as 71 percent of the participants followed an academic track and the study was conducted in a [details removed for peer review], possibly limiting the generalizability of the findings. That is, compared to an American context where the choice of a major occurs once enrolled in college or university, or to a German context where students have to apply for a major, in the [details removed for peer review], the study choice process takes place in the last year of secondary education and the enrollment at higher education institutions is (with some exceptions) unconditional, meaning without entrance exam, regardless of the chosen high school program and without a letter of admission or rejection. To add to the selectiveness, as Schwartz et al. (2005, p. 205) argued, there is a "forgotten half of emerging adults who do not attend college". It is important to examine whether similar associations between identity, parental involvement, and career decision-making w be observed among students who directly embark on a vocation after graduation from secondary education.

Conclusion

Because of the multiple social and academic stressors freshmen students encounter, a substantial group of students drops out by the end of their first year in higher education (Credé & Niehorster, 2012). As a clear, deeply explored, and autonomously regulated study choice might help freshmen students to cope with these stressors and to be successful in higher education (e.g., Germeijs & Verschueren, 2007), it is imperative to gain insight in the precursors of high-quality educational decision-making. Late adolescents who felt they were in touch with a sense of deeply held values, interests, and preferences were found to be more engaged in this study choice process and to more easily arrive at personally meaningful and intrinsically motivated plans for the future. To a lesser extent, the experience of having parents who are involved in need supportive ways also contributed to beter decision-making. Conversely, when parents were perceived as more controlling or uninvolved, late adolescents reported feeling more pressured to choose a study or to go through the motions of the study choice process more superficially.

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Table 1

Descriptives of and Correlations between the Study Variables

	1	2	3	4	5	6	7	8	9	10	11	12	13
Main predictors													
1. Need support mother	-												
2. Control mother	-.31**	-											
3. Uninvolvement mother	-.59**	.24*	-										
4. Need support father	.34**	-.19*	-.18*	-									
5. Control father	-.07	.37**	.21*	.12	-								
6. Uninvolvement father	-.32**	.24*	.41**	-.56**	.11	-							
7. Authentic Inner compass	.26**	-.07	-.29**	.12	-.18	-.20*	-						
Outcomes													
8. Autonomous motivation	.33**	-.15	-.28**	.17*	-.14	-.23*	.46**	-					
9. Controlled motivation	-.11	.19*	.21*	-.05	.24**	.27**	-.18*	-.21*	-				
10. Orientation	.39**	.02	-.16	.20*	.08	-.10	.23*	.38**	.13	-			
11. Exploration	.41**	-.06	-.25**	.28**	-.02	-.14	.34**	.41**	-.03	.53**	-		
12. Information	.31**	-.18	-.29**	.17*	-.20*	-.24**	.59**	.45**	-.12	.43**	.55**	-	
13. Commitment	.15	-.13	-.23*	.13	-.24**	-.25**	.49**	.49**	-.30**	.10	.33**	.48**	-
M	3.84	2.23	1.49	3.38	2.02	1.88	3.87	4.27	2.09	3.86	2.82	3.98	4.62
SD	0.78	0.82	0.60	0.94	0.88	0.82	0.60	0.70	0.91	0.73	0.55	0.59	0.95

Note. * $p < .01$, ** $p < .001$ (two-tailed). Correlational analyses were performed controlling for adolescents' gender and educational track.

Table 2

Effects of Common and Differential Parental Involvement and Adolescents' AIC to Study Choice Decision Process

	Autonomous motivation		Controlled motivation		Orientation to choice		Exploration		Information		Commitment	
	1	2	1	2	1	2	1	2	1	2	1	2
Common need support	.15	.16	.05	.11	.28*	.20	.45***	.55***	.10	.02	-.05	-.09
Differential need support	.18		-.09		.06		-.10		.16		.11	
Common control	-.09	-.14	.21*	.33**	.10	.18	.02	.00	-.12	-.15	-.19*	-.24*
Differential control	.03		-.08		.07		.15		.17		.10	
Common uninvolvement	-.17	-.10	.25*	.41*	.01	.06	.02	.31	-.23	-.17	-.24	-.30
Differential uninvolvement	.15		-.09		.04		-.22		.19		.22	
Authentic inner compass		.38***		-.12*		.23***		.33***		.49***		.36***
Common need support x AIC		.20		-.12		.11		.25		.19		.08
Common control x AIC		-.14		-.10		.02		.02		-.01		.11
Common uninvolvement x AIC		.33		-.20		.11		.15		.27		.06
R^2	.13**		.13*		.08*		.19**		.15**		.14**	

Note. * $p < .01$, ** $p < .001$. In the second analyses (step 2), R^2 was not offered in the output. Analyses were performed controlling for adolescents' gender and educational track.

The How and the Why of Study Choice Processes in Higher Education:

The Role of Parental Involvement and the Experience of Having an Authentic Inner Compass

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The present study shows first that need-supportive parental involvement in adolescents' study choice relates to a more adaptive study choice process, whereas controlling parental involvement and uninvolvement relate to a less adaptive decision-making process. Second, mothers are more involved than fathers in late adolescents' study choices, for better and for worse. Third, late adolescents experiencing a stronger authentic inner compass display a more adaptive decision-making process.